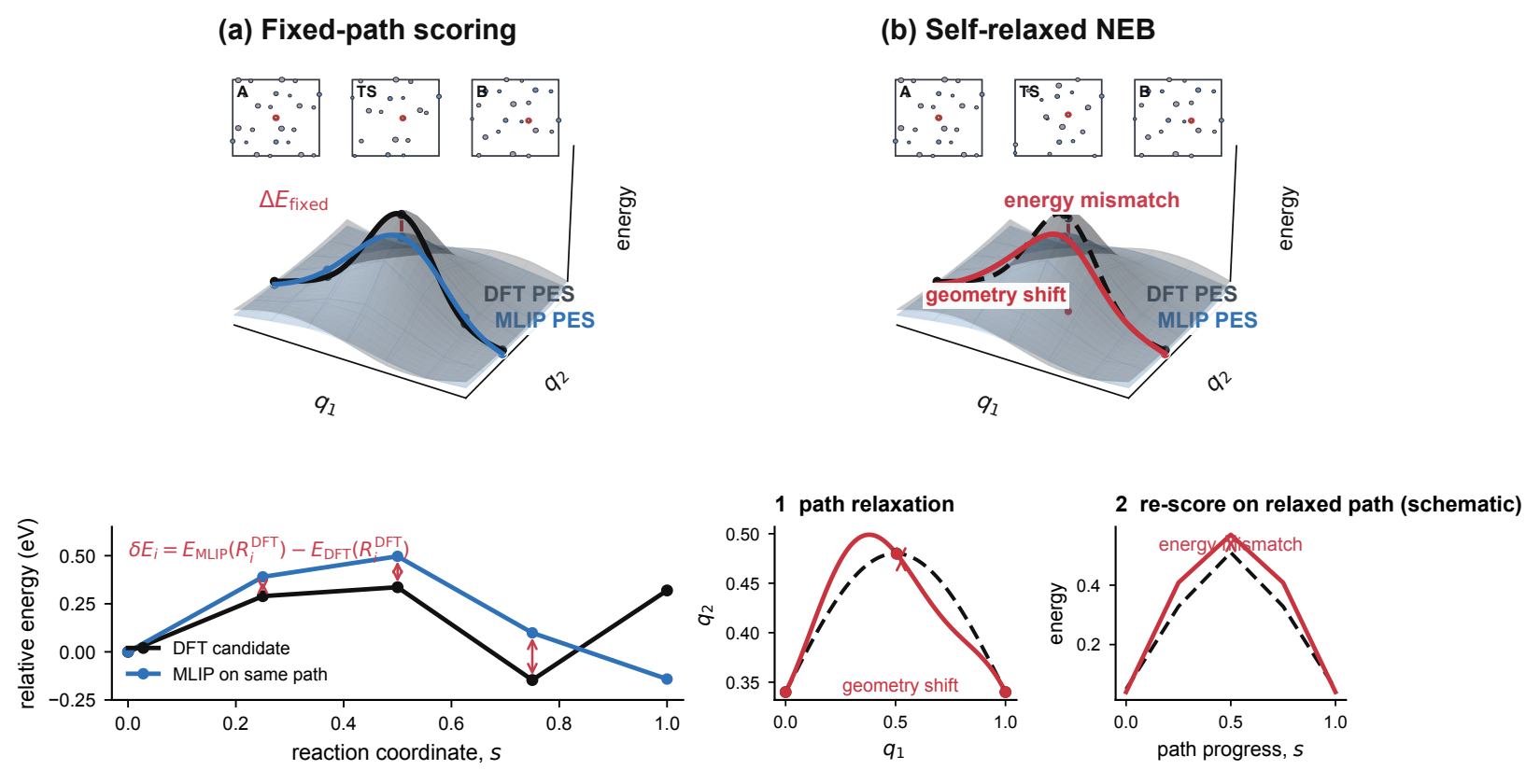

(a) Fixed-path scoring
A
TS
B
ΔE_{fixed}
DFT PES
MLIP PES
energy
q_1
q_2
(b) Self-relaxed NEB
A
TS
B
energy mismatch
geometry shift
DFT PES
MLIP PES
energy
q_1
q_2
$\delta E_i = E_{\text{MLIP}}(R_i^{\text{DFT}}) - E_{\text{DFT}}(R_i^{\text{DFT}})$
relative energy (eV)
0.50
0.25
0.00
−0.25
DFT candidate
MLIP on same path
0.0
0.2
0.4
0.6
0.8
1.0
reaction coordinate, s
1 path relaxation
0.50
0.45
0.40
0.35
q_2
geometry shift
0.0
0.5
1.0
q_1
2 re-score on relaxed path (schematic)
energy mismatch
energy
0.4
0.2
0.0
0.5
1.0
path progress, s